Arten

Edge Object Detection, Counting & Attributes

Judge brief · Offline edge AI for safer streets

Namma AI Hackathon 2026 · Problem Statement 2 · [Team name] · [College / organisation]

One sentence

Arten counts pedestrians and vehicles offline on a sub-₹10,000 Raspberry Pi, so a school can justify a crossing guard without buying a surveillance system.

The problem

- Road traffic injuries are the **leading cause of death for children and young adults aged 5–29** (WHO).
- India recorded **1,68,491 road deaths in 2022**; pedestrians were **19.5%**, two-wheeler riders **44.5%** (MoRTH).
- The missing data is simple: how many people cross, when, and alongside how much traffic. That data is usually locked behind expensive cameras or cloud subscriptions.

What it does

1. Takes a **10–20 second clip** or **10–20 images**.
2. Detects **person, bicycle, car, motorcycle, bus, truck**.
3. Prints **per-frame counts** and writes `counts.csv` / `detections.jsonl`.
4. Saves **annotated frames, a contact sheet and an optional annotated video**.
5. Reports **p50/p95 latency, throughput, model size and memory**.
6. Adds **on-device attributes**: image-plane direction, dwell, relative speed and optional coarse vehicle colour.
7. Runs entirely locally: **no cloud, no identities**.

Numbers that matter

Metric	Value	Basis
Model	YOLO11n INT8, 4.73 MB	Arm-optimized ExecuTorch `.pte`
Pi 5 reference latency	134.10 ms p50 / 7.46 FPS	Arm model card
Pi 5 reference memory	59.17 MB peak	Arm model card
COCO accuracy	38.90% mAP@0.5:0.95	Arm model card
Dev-host model latency	20.34 ms mean / 49.16 FPS	This repo, labelled dev
Sample clip	150 frames, peak 9 people/frame, 18 tracked people	This repo, indicative
Clip attributes	9 left / 4 right / 2 stationary / 2 down / 1 up; mean dwell 4.5 s	This repo, derived
Traffic colours	17 cars: 9 white, 4 gray/silver, 2 blue, 1 red, 1 orange/brown	This repo, HSV heuristic
Tests	29 passing	This repo

Mandatory deliverables

Requirement	Evidence
Clip/images on ExecuTorch + Pi	`data/samples/pedestrians_15s.mp4` ; `scripts/setup_pi.sh`
Per-frame counts	`results/demo_pedestrians/counts.csv` ; HUD
Category + counts + 2 samples	Contact sheet + two annotated Bengaluru images
Latency, model, throughput	`summary.json` ; `benchmark_dev.json`
2-minute explanation	`docs/pitch-2min.md`

Why it is credible

- The model is a **target-verified Arm artifact**, not a laptop script.
- Every stage is timed: preprocess, inference, decode/NMS, tracking, annotation.
- The report separates **Pi reference** from **dev-host measurement**.
- The COCO/India gap is named: auto-rickshaws are not a class; dense-scene motorcycles are missed.
- The attribute layer is **derived, not learned**: image-plane direction, uncalibrated speed, coarse HSV colour. The caveats are printed with every run.
- The repo ships **29 tests**, a one-command Pi setup, a checksum-verified model and a reproducible evidence bundle.

Practical application

A school records three 15-second clips: morning drop-off, afternoon dismissal and a quiet hour. It runs them on the Pi, offline, and compares peak pedestrian counts, vehicle counts and observed person-seconds. A high pedestrian/vehicle overlap at dismissal, plus direction and dwell attributes, is evidence for a crossing guard, a zebra crossing or a no-parking zone at the gate. One Pi. No subscription.

Demo commands

```
arten doctor
arten run -i data/samples/pedestrians_15s.mp4 \
  -o results/demo --save-video --estimate-vehicle-colors
arten benchmark --input data/samples/bengaluru_traffic.jpg --runs 50
```

Links

- GitHub: https://github.com/sujalmh/arten-edge-ai
- Proposal: `docs/proposal.md`
- Technical report: `docs/report.md`
- Two-minute pitch: `docs/pitch-2min.md`
- Demo video: `results/demo.mp4`

Count movement, not people.